

Adam Ingah

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Summary

Electronics Engineer with experience in embedded systems, FPGA Design, and hardware development. U.S. Citizen based in Seattle, WA.

Education

University of Washington, Seattle M.S. Electrical Engineering June 2027

- Relevant Coursework: Digital Design with FPGAs, Embedded Systems Programming

University of Alabama, Birmingham B.S. Electrical Engineering May 2025

- Relevant Coursework: Machine Learning with Python, Signals and Systems

Experience

Hardware Lead | UAB Engineering Lab | 8/2023 - 5/2024

- Designed and prototyped a low-cost assistive vision device for visually impaired users, reducing cost by ~90% compared to commercial solutions
- Led hardware testing and iterative prototyping in partnership with engineers and patients, refining usability and accuracy through feedback-driven design cycles.

Lab Assistant | UAB Libraries | 8/2022 - 5/2025

- Maintained and repaired 5+ 3D printers, enabling 20+ student projects throughout the semester with minimal printer downtime.
- Collaborated with student engineers and researchers during design reviews, offering technical troubleshooting and CAD guidance to improve project outcomes.

Projects

Autonomous Fire Extinguishing Robot | adamingah.com/projects/pages/amfr.html

- Built an autonomous robot using Raspberry Pi 4 and thermal vision to detect and extinguish fires in real time
- Led hardware integration and rapid prototyping with 3D-printed components, ensuring reliable sensor-actuator communication under test conditions.

IoT Plant Irrigation System | adamingah.com/projects/pages/ipws.html

- Developed an ESP32-based irrigation system with real-time soil monitoring and automated pump control, reducing manual watering needs.
- Integrated analog sensors with Arduino IoT Cloud via UART/SPI/I2C protocols, enabling remote data tracking and control.

Low Vision Magnifier | adamingah.com/projects/pages/lvm.html

- Designed a text magnifier with 3D-printed housings and Raspberry Pi-based OCR (OpenCV + Tesseract) for individuals with low vision.
- Delivered an affordable, modular, and accessible solution compared to commercial devices.

Skills

Programming: C, C++, Assembly, Java, Python, SystemVerilog, MATLAB

Hardware: ESP32, Arduino, Raspberry Pi,

Tools: LTSpice, Altium Designer, Git, SolidWorks, Fusion 360